← Back **Assessment 6**
Graded Assignment • 15 min

Due Aug 13, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. What is the first step in manually dividing an image dataset into training and testing datasets? 1 / 1 point

- ( ) Applying data augmentation techniques.
- ( ) Using a pre-trained model for initial insights.
- (•) Collecting and organizing the entire dataset.
- ( ) Training the model with initial data.

Correct
Correct! Before splitting the dataset, you need to have the dataset collected and organized.

2. Why is it important to randomly allocate images to training and testing datasets? 1 / 1 point

- ( ) To reduce the size of the dataset.
- ( ) To easily convert images to grayscale.
- ( ) To speed up the training process.
- (•) To prevent overfitting and ensure the model generalizes well.

Correct
Correct! Random allocation helps to prevent overfitting and makes sure the model generalizes well to new data.

3. How should images be allocated to training and testing directories to achieve a 70/30 split? 1 / 1 point

- ( ) 60% of images to the training directory, 40% to the testing directory.
- ( ) 50% of images to the training directory, 50% to the testing directory.
- (•) 70% of images to the training directory, 30% to the testing directory.
- ( ) 80% of images to the training directory, 20% to the testing directory.

Correct
Correct! This is the standard approach for a 70/30 split.

4. What is the primary function of a convolutional layer in a CNN? 1 / 1 point

- ( ) To increase the number of channels in the input image.
- ( ) To reduce the spatial dimensions of the input image.
- (•) To detect and extract features from the input image.
- ( ) To flatten the input image into a single vector.

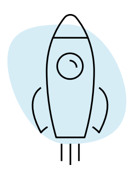
Correct
Correct! The convolutional layer applies filters to the input image to detect and extract features.

5. What is the purpose of using the 'flow from directory' method in image data generation? 1 / 1 point

- ( ) To reduce the image resolution for faster training.
- ( ) To automatically balance the number of images in each class.
- (•) To load images directly from a directory for preprocessing and augmentation.
- ( ) To convert images to grayscale before training a model.

Correct
Correct! The 'flow from directory' method is used to load images from a directory, which are then preprocessed and augmented.

Close ✕



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